CUSTOMER 360° & AI-BASED SEGMENTATION ANALYSIS

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DATE: JAN 2026

1. WHAT PROBLEM WAS THIS PROJECT TRYING TO SOLVE?

Businesses often collect large amounts of customer transaction data, but it can be difficult to understand customer purchasing behavior from raw datasets. Without proper analysis, companies cannot easily identify their most valuable customers or understand purchasing trends.

The goal of this project was to transform raw e-commerce transaction data into meaningful insights using Power BI. By creating an interactive dashboard, businesses can analyze customer activity, revenue contribution, and order behavior more effectively.

2. WHAT DATASET WAS USED FOR THIS PROJECT?

This project used an E-commerce Retail Dataset containing customer transaction data.

The dataset includes key information such as:

InvoiceNo
StockCode / Product code
Description
Quantity
InvoiceDate
UnitPrice
CustomerID
Country

After cleaning the data, the dataset contained more than 3,000 valid transaction records used for analysis.

3. HOW WAS THE DATA PREPARED FOR ANALYSIS?

Data cleaning and transformation were performed using Power Query in Power BI.

The main data preparation steps included:

- Removing rows with missing CustomerID
- Filtering out negative quantities (product returns)
- Removing invalid price values
- Correcting data types for numeric and date columns
- Creating a calculated column called Revenue

Revenue was calculated using the formula:

Revenue = Quantity * UnitPrice

These transformations ensured the dataset was accurate and ready for analytical processing.

4. HOW WERE BUSINESS METRICS CALCULATED?

Key performance indicators (KPIs) were created using DAX (Data Analysis Expressions) to measure customer and revenue performance. Examples of measures created include:

Total Revenue

```
Total Revenue = SUM(Data[Revenue])
```

Total Customers

```
Total Customers = DISTINCTCOUNT(Data[CustomerID])
```

Total Orders

```
Total Orders = DISTINCTCOUNT(Data[InvoiceNo])
```

Average Order Value

```
Avg Order Value = DIVIDE([Total Revenue], [Total Orders])
```

These measures provide important insights into overall customer activity and sales performance.

5. WHAT SOLUTION WAS DEVELOPED?

An interactive Customer Analytics Dashboard was developed using Power BI to visualize the key insights from the dataset.
The dashboard includes several visualizations designed to highlight important customer and revenue patterns.

6. DASHBOARD VISUALIZATIONS

The dashboard includes the following components:
KPI Cards
Four key metrics are displayed at the top of the dashboard:

- Total Revenue
- Total Customers
- Total Orders
- Average Order Value

These KPIs provide a quick overview of business performance.

Revenue Trend (Line Chart)

This chart shows how revenue changes over time, helping identify growth patterns and seasonal trends.

Top 10 Customers (Clustered Bar Chart)

This visualization highlights the top customers based on total revenue, helping identify the most valuable customers.

Customers by Country (Clustered Bar Chart)

This chart shows the distribution of customers across different countries, allowing businesses to understand their geographical customer base.

Customer Detail Table

A detailed table was included to display customer-level metrics such as revenue, order count, and average order value.
This allows users to explore detailed customer information directly from the dashboard.

7. INTERACTIVE FEATURES

To improve usability, the dashboard includes interactive filters (slicers):

- Country filter
- Year filter

These slicers allow users to dynamically explore the data and focus on specific regions or time periods.

8. KEY INSIGHTS FROM THE ANALYSIS

From the dashboard, several insights can be observed:

- A small group of customers contributes a large share of the total revenue.
- Customer distribution varies across countries, highlighting key markets.
- Some customers place frequent orders, indicating loyal purchasing behavior.
- Revenue trends change over time, suggesting seasonal patterns in purchasing.

These insights can help businesses improve marketing strategies and customer targeting.

9. TOOLS AND TECHNOLOGIES USED

This project was developed using the following tools:

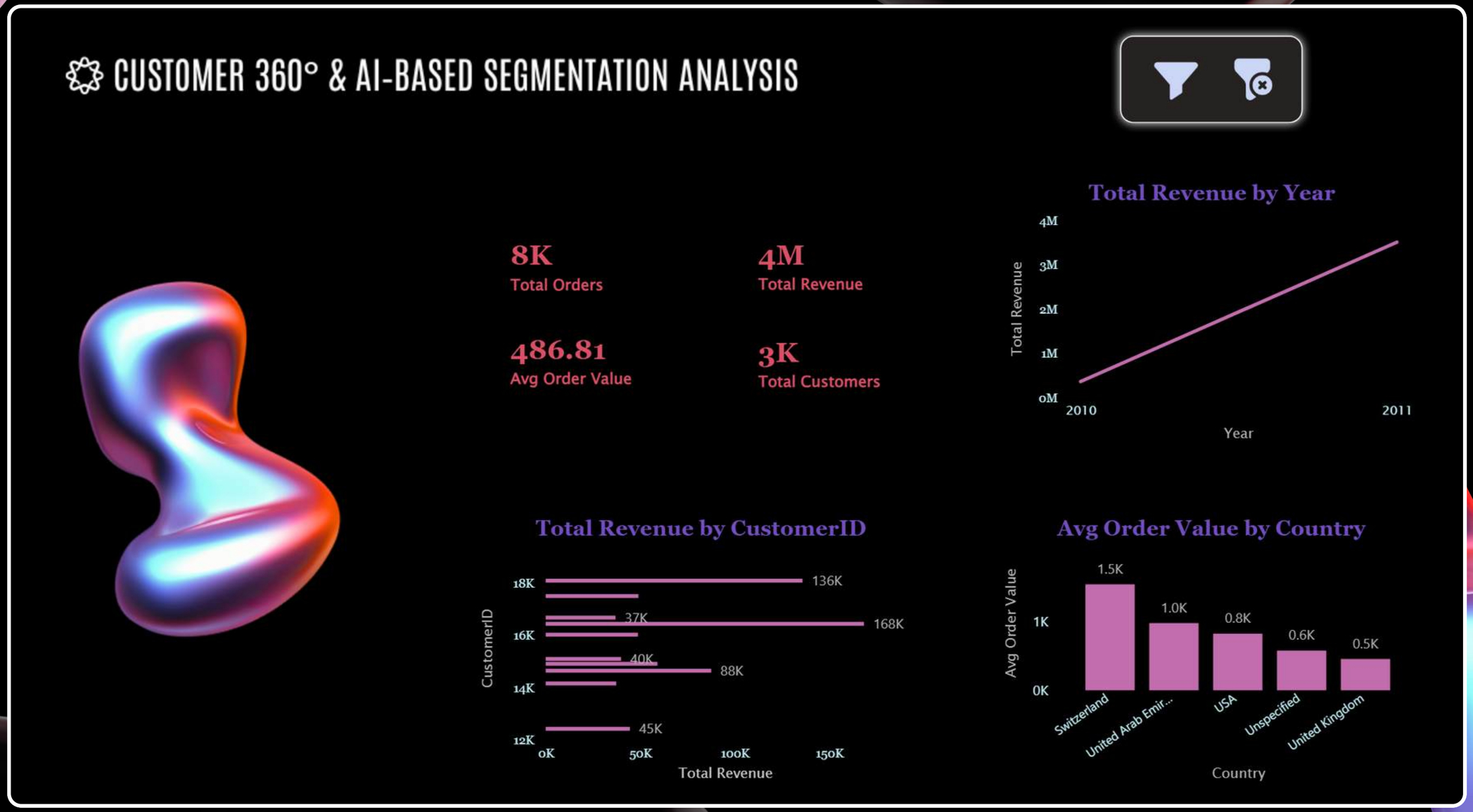
- Power BI for dashboard development and visualization
- Power Query for data cleaning and transformation
- DAX for calculating business metrics
- GitHub for project portfolio management

10. CONCLUSION

This project demonstrates how Power BI can be used to analyze customer transaction data and transform it into meaningful insights.

By combining data cleaning, DAX calculations, and interactive visualizations, the dashboard provides a clear view of customer behavior and revenue performance. Such dashboards help organizations better understand their customers and make more informed business decisions.

DASHBOARD PREVIEW



THANK YOU FOR YOUR ATTENTION

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